Euler's Method and Logistic Growth

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Euler's Method Iterations**

A population of bacteria follows a logistic growth model where $P(t)$ is the population at time t (in hours). The differential equation governing the population growth is $\frac{dP}{dt} = 0.05P\left(1 - \frac{P}{1000}\right)$. If the initial population is $P(0) = 200$, use Euler's method with a step size of $\Delta t = 1$ hour to approximate the population after 1 hour, $P(1)$.

Euler's method approximates the next value $P(t + \Delta t)$ using the formula $P(t + \Delta t) \approx P(t) + \frac{dP}{dt}|_t \cdot \Delta t$. We are given $P(0) = 200$ and $\Delta t = 1$. First, we find the rate of change at $t = 0$: $\frac{dP}{dt}|_{t=0} = 0.05 \cdot P(0)\left(1 - \frac{P(0)}{1000}\right)$ $\frac{dP}{dt}|_{t=0} = 0.05 \cdot 200\left(1 - \frac{200}{1000}\right)$ $\frac{dP}{dt}|_{t=0} = 10(1 - 0.2)$ $\frac{dP}{dt}|_{t=0} = 10 \cdot 0.8$ $\frac{dP}{dt}|_{t=0} = 8$ Now, we use Euler's method to approximate $P(1)$: $P(1) \approx P(0) + \frac{dP}{dt}|_{t=0} \cdot \Delta t$ $P(1) \approx 200 + 8 \cdot 1$ $P(1) \approx 208$

2 **[Foundational]** • **Target: The Logistic Differential Equation**

The graph of a differentiable function f is shown below. The function f has a local maximum at $x = a$ and a local minimum at $x = b$. The tangent line to the graph of f at $x = c$ has the equation $y - f(c) = \frac{1}{2}(x - c)$. Given that $f(c) = 3$ and $f'(c) = \frac{1}{2}$, find the value of c.

The equation of the tangent line to the graph of a function f at $x = c$ is given by $y - f(c) = f'(c)(x - c)$. We are given that the equation of the tangent line at $x = c$ is $y - f(c) = \frac{1}{2}(x - c)$. Comparing this to the general form, we see that $f'(c) = \frac{1}{2}$. We are also given that $f(c) = 3$. The graph shows a point on the function $f(x)$ at $(x, y) = (2, 3)$. Since the point $(c, f(c))$ must lie on the graph of f, we have $c = 2$ and $f(c) = 3$. Therefore, the value of c is 2.

3 **[Foundational]** • **Target: Analytical Solutions to Logistic Models**

A particle's position in the xy-plane is given by the parametric equations $x(t) = at^2 - 3t$ and $y(t) = 2t^3 + bt$, where a and b are constants. At time $t = 1$, the particle's velocity vector is $\mathbf{v}(1) = \langle 9, 10 \rangle$. Find the value of a and the value of b.

The velocity vector is given by the derivatives of the position components with respect to time: $x'(t) = \frac{dx}{dt} = 2at - 3$ and $y'(t) = \frac{dy}{dt} = 6t^2 + b$. We are given that at $t = 1$, the velocity vector is $\mathbf{v}(1) = \langle 9, 10 \rangle$. Therefore, we can set up the following system of equations: For the x-component: $x'(1) = 2a(1) - 3 = 9$ $2a - 3 = 9$ $2a = 12$ $a = 6$ For the y-component: $y'(1) = 6(1)^2 + b = 10$ $6 + b = 10$ $b = 4$ Thus, the values are $a = 6$ and $b = 4$.

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4 [Foundational] • Target: Advanced Slope Fields with Autonomous Equations

Evaluate $\lim_{x\to 0}\frac{\tan x}{x}$.

Method 1 · Trigonometric Limit Decomposition: Express $\tan x$ in terms of sine and cosine: $\frac{\tan x}{x}=\frac{\sin x}{x\cos x}=\left(\frac{\sin x}{x}\right)\cdot\left(\frac{1}{\cos x}\right)$. Applying limit product laws and the fundamental trigonometric limit $\lim_{x\to 0}\frac{\sin x}{x}=1$: $\lim_{x\to 0}\frac{\tan x}{x}=\left(\lim_{x\to 0}\frac{\sin x}{x}\right)\cdot\left(\lim_{x\to 0}\frac{1}{\cos x}\right)=(1)\cdot\left(\frac{1}{\cos 0}\right)=(1)\cdot(1)=\mathbf{1}$. Method 2 · L'Hôpital's Rule (Indeterminate Form $\frac{0}{0}$): Direct substitution yields $\frac{\tan 0}{0}=\frac{0}{0}$. Since both numerator and denominator are differentiable near $x=0$: $\lim_{x\to 0}\frac{\tan x}{x}=\lim_{x\to 0}\frac{\frac{d}{dx}[\tan x]}{\frac{d}{dx}[x]}=\lim_{x\to 0}\frac{\sec^2 x}{1}=\sec^2(0)=1$.

5 [Foundational] • Target: Euler's Method Iterations

For $x=\cos t,\ y=\sin t$, find $\frac{dy}{dx}$ at $t=\frac{\pi}{2}$.

Theorem (Parametric Derivatives): If $x(t)$ and $y(t)$ are differentiable functions of t and $x'(t)e0$, the first derivative of y with respect to x is given by: $\frac{dy}{dx}=\frac{y'(t)}{x'(t)}=\frac{dy/dt}{dx/dt}$. Step 1 · Differentiate both parametric components with respect to t: $\frac{dx}{dt}=\frac{d}{dt}[\cos t]=-\sin t,\quad \frac{dy}{dt}=\frac{d}{dt}[\sin t]=\cos t$. Step 2 · Form the derivative ratio: $\frac{dy}{dx}=\frac{\cos t}{-\sin t}=-\cot t$. Step 3 · Evaluate at $t=\frac{\pi}{2}$: $\left.\frac{dy}{dx}\right|_{t=\pi/2}=-\cot\left(\frac{\pi}{2}\right)=-\frac{\cos(\pi/2)}{\sin(\pi/2)}=-\frac{0}{1}=\mathbf{0}$. Geometric Interpretation: At $t=\frac{\pi}{2}$, the point on the unit circle is $(x,y)=(0,1)$. Since $x'(\pi/2)=-1e0$ and $y'(\pi/2)=0$, the curve has a horizontal tangent line $y=1$.

6 [Foundational] • Target: The Logistic Differential Equation

Find the area enclosed by the polar curve $r=2$.

Theorem (Area Bounded by a Polar Curve): The area A enclosed by a continuous polar function $r(\theta)$ on the interval $[\alpha,\beta]$ is given by: $A=\frac{1}{2}\int_{\alpha}^{\beta}[r(\theta)]^2\,d\theta$. Step 1 · Establish domain of integration: The equation $r=2$ describes a circle of constant radius $R=2$ centered at the origin (pole). A complete revolution traces the circle exactly once over $\theta\in[0,2\pi]$. Step 2 · Set up and evaluate the definite polar integral: $A=\frac{1}{2}\int_0^{2\pi}(2)^2\,d\theta=\frac{1}{2}\int_0^{2\pi}4\,d\theta=2\int_0^{2\pi}1\,d\theta=2[\theta]_0^{2\pi}=2(2\pi-0)=\mathbf{4\pi}$. Verification: By Euclidean geometry, the area of a circle with radius $R=2$ is $A=\pi R^2=\pi(2)^2=4\pi$, in exact agreement.

7 [Foundational] • Target: Analytical Solutions to Logistic Models

Give the first three nonzero terms of the Maclaurin series for $\sin x$.

Theorem (Maclaurin Series for Sine): The sine function is odd, so its Maclaurin series contains exclusively odd powers of x with alternating signs. Step 1 · Compute successive derivatives at $x=0$: $f(x)=\sin x\Rightarrow f(0)=0\ f'(x)=\cos x\Rightarrow f'(0)=1\ f''(x)=-\sin x\Rightarrow f''(0)=0\ f'''(x)=-\cos x\Rightarrow f'''(0)=-1$ $f^{(4)}(x)=\sin x\Rightarrow f^{(4)}(0)=0\ f^{(5)}(x)=\cos x\Rightarrow f^{(5)}(0)=1$ Step 2 · Form the first three nonzero terms: $\frac{f'(0)}{1!}x+\frac{f'''(0)}{3!}x^3+\frac{f^{(5)}(0)}{5!}x^5=\mathbf{x-\frac{x^3}{3!}+\frac{x^5}{5!}=x-\frac{x^3}{6}+\frac{x^5}{120}}$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Advanced Slope Fields with Autonomous Equations

Give the first three terms of the Maclaurin series for $\cos x$.

Theorem (Maclaurin Series for Cosine): The cosine function is even, so its Maclaurin series consists exclusively of even powers of x with alternating signs. Step 1 · General Maclaurin formula for $\cos x$: $\cos x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!}$. Step 2 · Write out the terms for $n = 0, 1, 2$: $n = 0 \Rightarrow (-1)^0 \frac{x^0}{0!} = 1$ $n = 1 \Rightarrow (-1)^1 \frac{x^2}{2!} = -\frac{x^2}{2}$ $n = 2 \Rightarrow (-1)^2 \frac{x^4}{4!} = +\frac{x^4}{24}$ Conclusion: The first three terms are: $\mathbf{1 - \frac{x^2}{2!} + \frac{x^4}{4!} = 1 - \frac{x^2}{2} + \frac{x^4}{24}}$.

9 [Intermediate] • Target: Euler's Method Iterations

The graph of the function $f(x) = x^2 - 4x + 3$ is shown below. What is the definite integral of $f(x)$ from $x = 1$ to $x = 3$?

To find the definite integral of $f(x) = x^2 - 4x + 3$ from $x = 1$ to $x = 3$, we evaluate $\int_1^3 (x^2 - 4x + 3)\,dx$. First, find the antiderivative of $f(x)$: $F(x) = \frac{x^3}{3} - \frac{4x^2}{2} + 3x = \frac{x^3}{3} - 2x^2 + 3x$. Next, apply the Fundamental Theorem of Calculus: $F(3) - F(1)$. Calculate $F(3)$: $F(3) = \frac{3^3}{3} - 2(3)^2 + 3(3) = \frac{27}{3} - 2(9) + 9 = 9 - 18 + 9 = 0$. Calculate $F(1)$: $F(1) = \frac{1^3}{3} - 2(1)^2 + 3(1) = \frac{1}{3} - 2 + 3 = \frac{1}{3} + 1 = \frac{4}{3}$. Finally, subtract $F(1)$ from $F(3)$: $F(3) - F(1) = 0 - \frac{4}{3} = -\frac{4}{3}$.

10 [Intermediate] • Target: The Logistic Differential Equation

Consider the region bounded by the curve $y = \frac{1}{x^2+1}$, the x-axis, and the vertical lines $x = 0$ and $x = b$, where $b > 0$. If the area of this region is $\frac{\pi}{3}$, what is the value of b?

The area of the region is given by the definite integral of the function from $x = 0$ to $x = b$. The integral of $\frac{1}{x^2+1}$ is $\arctan(x)$. Therefore, the area is given by: Area $= \int_0^b \frac{1}{x^2+1} dx = [\arctan(x)]_0^b = \arctan(b) - \arctan(0) = \arctan(b)$ We are given that the area is $\frac{\pi}{3}$. So, we have: $\arctan(b) = \frac{\pi}{3}$ To find b, we take the tangent of both sides: $b = \tan\left(\frac{\pi}{3}\right)$ We know that $\tan\left(\frac{\pi}{3}\right) = \sqrt{3}$. Therefore, $b = \sqrt{3}$.

11 [Intermediate] • Target: Analytical Solutions to Logistic Models

Evaluate $\int_0^2 6x^2\,dx$.

Theorem (Fundamental Theorem of Calculus, Part 2): If f is continuous on $[a, b]$ and F is any antiderivative of f such that $F'(x) = f(x)$, then: $\int_a^b f(x)\,dx = F(b) - F(a) = [F(x)]_a^b$. Step 1 · Find the indefinite antiderivative $F(x)$: Using the power rule of integration $\int x^n\,dx = \frac{x^{n+1}}{n+1} + C$: $F(x) = \int 6x^2\,dx = 6 \cdot \frac{x^3}{3} = 2x^3$. Step 2 · Evaluate at the upper and lower limits: $\int_0^2 6x^2\,dx = [2x^3]_0^2 = 2(2)^3 - 2(0)^3 = 2(8) - 0 = \mathbf{16}$.

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12 [Intermediate] • Target: Advanced Slope Fields with Autonomous Equations

A point is represented in polar coordinates by $(r, \theta) = (5, \frac{\pi}{4})$. What are the equivalent Cartesian coordinates (x, y)?

The conversion formulas from polar coordinates (r, θ) to Cartesian coordinates (x, y) are $x = r\cos(\theta)$ and $y = r\sin(\theta)$. Given $r = 5$ and $\theta = \frac{\pi}{4}$, we have: $x = 5\cos(\frac{\pi}{4}) = 5 \cdot \frac{\sqrt{2}}{2} = \frac{5\sqrt{2}}{2}$ $y = 5\sin(\frac{\pi}{4}) = 5 \cdot \frac{\sqrt{2}}{2} = \frac{5\sqrt{2}}{2}$ Therefore, the Cartesian coordinates are $(\frac{5\sqrt{2}}{2}, \frac{5\sqrt{2}}{2})$.

13 [Intermediate] • Target: Euler's Method Iterations

Evaluate $\int x \sin x \, dx$.

Method · Integration by Parts (LIATE Selection): Step 1 · Assign u and dv: Algebraic (x) precedes Trigonometric ($\sin x$), so set: $u = x \Rightarrow du = dx, \quad dv = \sin x \, dx \Rightarrow v = -\cos x$. Step 2 · Apply $\int u \, dv = uv - \int v \, du$: $\int x \sin x \, dx = x(-\cos x) - \int (-\cos x) \, dx = -x\cos x + \int \cos x \, dx$. Step 3 · Integrate: $-x\cos x + \sin x + C = \boldsymbol{\sin \mathbf{x} - \mathbf{x}\cos \mathbf{x} + \mathbf{C}}$.

14 [Intermediate] • Target: The Logistic Differential Equation

A region in the first quadrant is bounded by the graph of $f(x) = 3x^2 + 1$, the x-axis, and the vertical lines $x = 1$ and $x = 2$. What is the area of this region?

The area of the region is given by the definite integral of $f(x)$ from $x = 1$ to $x = 2$. Area $= \int_1^2 (3x^2 + 1) \, dx$ First, find the antiderivative of $3x^2 + 1$: Antiderivative $= \frac{3x^3}{3} + x = x^3 + x$ Now, evaluate the definite integral using the Fundamental Theorem of Calculus: Area $= [x^3 + x]_1^2$ Area $= (2^3 + 2) - (1^3 + 1)$ Area $= (8 + 2) - (1 + 1)$ Area $= 10 - 2$ Area $= 8$

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Analytical Solutions to Logistic Models • Cross-Lesson: 5.01 Sequences and Basic Series Convergence Tests

Consider the function $f(x) = x^2 - 6x + 10$. Find the x-coordinates of the points on the graph of $y = f(x)$ where the tangent line is horizontal.

A horizontal tangent line occurs where the derivative of the function is equal to zero. First, find the derivative of $f(x)$: $f'(x) = \frac{d}{dx}(x^2 - 6x + 10) = 2x - 6$. Next, set the derivative equal to zero and solve for x : $2x - 6 = 0$. Adding 6 to both sides gives $2x = 6$. Dividing by 2 gives $x = 3$. Therefore, the x-coordinate of the point where the tangent line is horizontal is $x = 3$.

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16 **[Challenging]** • **Target: Advanced Slope Fields with Autonomous Equations** • **Cross-Lesson: 5.02 Ratio, Root, and Alternating Series Tests**

For what x does $\sum_{n=0}^{\infty} \frac{x^n}{n!}$ converge?

Theorem (Ratio Test for Absolute Convergence): For an infinite series $\sum a_n$, evaluate $L = \lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right|$. The series converges absolutely when $L < 1$. Step 1 · Set up the ratio for $a_n = \frac{x^n}{n!}$: $L = \lim_{n\to\infty} \left|\frac{x^{n+1}}{(n+1)!} \cdot \frac{n!}{x^n}\right| = \lim_{n\to\infty} \left|\frac{x \cdot x^n}{(n+1)\cdot n!} \cdot \frac{n!}{x^n}\right| = \lim_{n\to\infty} \frac{|x|}{n+1}$. Step 2 · Evaluate the limit for any real number x: For any fixed $x \in \mathbb{R}$, $|x|$ is a finite constant, whereas $n+1 \to \infty$: $L = |x| \lim_{n\to\infty} \frac{1}{n+1} = |x| \cdot 0 = 0$. Step 3 · Conclude interval and radius of convergence: Since $L = 0 < 1$ for every real value of x, the series converges absolutely for all real numbers: $\mathbf{x \in (-\infty, \infty)}$, $\text{Radius of Convergence: } \mathbf{R = \infty}$. Identity: This is the Maclaurin series representation of e^x.

17 **[Challenging]** • **Target: Euler's Method Iterations** • **Cross-Lesson: 6.01 Power Series and Interval of Convergence**

The polar curve $r = 2\cos\theta$ is a circle. Find its enclosed area.

Theorem (Area of Circle in Polar Coordinates): The curve $r = 2\cos\theta$ is a circle passing through the pole. Multiplying by r yields $r^2 = 2r\cos\theta \Rightarrow x^2 + y^2 = 2x \Rightarrow (x-1)^2 + y^2 = 1$, which is a circle of radius $R = 1$ centered at $(1, 0)$. Step 1 · Identify integration bounds: The circle is completely traced as θ goes from $-\frac{\pi}{2}$ to $\frac{\pi}{2}$ (or from 0 to π). Step 2 · Set up and evaluate the polar integral: $A = \frac{1}{2}\int_{-\pi/2}^{\pi/2} (2\cos\theta)^2\, d\theta = \frac{1}{2}\int_{-\pi/2}^{\pi/2} 4\cos^2\theta\, d\theta = 2\int_{-\pi/2}^{\pi/2} \cos^2\theta\, d\theta$. By symmetry: $A = 4\int_0^{\pi/2} \frac{1+\cos(2\theta)}{2}\, d\theta = 2\int_0^{\pi/2} (1+\cos(2\theta))\, d\theta = 2\left[\theta + \frac{1}{2}\sin(2\theta)\right]_0^{\pi/2}$. $A = 2\left(\frac{\pi}{2} + 0 - (0+0)\right) = \mathbf{\pi}$. Verification: By Cartesian geometry, radius is $R = 1 \Rightarrow A = \pi R^2 = \pi(1)^2 = \pi$.

18 **[Challenging]** • **Target: The Logistic Differential Equation** • **Cross-Lesson: 6.02 Taylor and Maclaurin Series**

Find the area enclosed by the cardioid $r = 1 + \cos\theta$.

Theorem (Area of a Cardioid): The cardioid $r = 1 + \cos\theta$ is symmetric about the polar axis (x-axis) and is completely traced as θ ranges from 0 to 2π. $A = \frac{1}{2}\int_0^{2\pi} [r(\theta)]^2\, d\theta$. Step 1 · Expand $[r(\theta)]^2$: $(1+\cos\theta)^2 = 1 + 2\cos\theta + \cos^2\theta$. Step 2 · Apply the trigonometric power-reduction identity $\cos^2\theta = \frac{1+\cos(2\theta)}{2}$: $1 + 2\cos\theta + \frac{1+\cos(2\theta)}{2} = \frac{3}{2} + 2\cos\theta + \frac{1}{2}\cos(2\theta)$. Step 3 · Integrate over $[0, 2\pi]$: $A = \frac{1}{2}\int_0^{2\pi} \left(\frac{3}{2} + 2\cos\theta + \frac{1}{2}\cos(2\theta)\right) d\theta = \frac{1}{2}\left[\frac{3}{2}\theta + 2\sin\theta + \frac{1}{4}\sin(2\theta)\right]_0^{2\pi}$. Since $\sin(2\pi) = \sin(4\pi) = \sin(0) = 0$: $A = \frac{1}{2}\left(\frac{3}{2}(2\pi) - 0\right) = \frac{1}{2}(3\pi) = \mathbf{\frac{3\pi}{2}}$.

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19 [Challenging] • Target: Analytical Solutions to Logistic Models • Cross-Lesson: 1.01 Integration by Parts

Consider the function $f(x)$ defined by its Maclaurin series $f(x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!\, 3^{2n+1}}$. Let $g(x) = \int_0^x f(t)dt$. The function $f(x)$ is a known elementary function. Determine the value of $g''(0)$.

The Maclaurin series for $f(x)$ is given by $f(x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!\, 3^{2n+1}}$. This is the Maclaurin series for $\sin(u)$ with $u = \frac{x}{3}$. Therefore, $f(x) = \sin\left(\frac{x}{3}\right)$. We are given $g(x) = \int_0^x f(t)dt$. To find $g''(0)$, we first find the derivatives of $g(x)$: First derivative: $g'(x) = \frac{d}{dx}\int_0^x f(t)dt = f(x)$ by the Fundamental Theorem of Calculus. Second derivative: $g''(x) = f'(x)$. We need to evaluate $g''(0)$, which is equal to $f'(0)$. We have $f(x) = \sin\left(\frac{x}{3}\right)$. So, $f'(x) = \frac{d}{dx}\sin\left(\frac{x}{3}\right) = \cos\left(\frac{x}{3}\right) \cdot \frac{1}{3} = \frac{1}{3}\cos\left(\frac{x}{3}\right)$. Now, evaluate $f'(0)$: $f'(0) = \frac{1}{3}\cos\left(\frac{0}{3}\right) = \frac{1}{3}\cos(0) = \frac{1}{3} \cdot 1 = \frac{1}{3}$. Therefore, $g''(0) = \frac{1}{3}$.

20 [Challenging] • Target: Advanced Slope Fields with Autonomous Equations • Cross-Lesson: 1.02 Partial Fractions and Improper Integrals

Evaluate $\int_1^{\infty} \frac{1}{x^2}\, dx$.

Theorem (Improper Integral with Infinite Limit): An improper integral with an infinite upper limit is formally defined as the limit of proper definite integrals: $\int_a^{\infty} f(x)\, dx = \lim_{b\to\infty} \int_a^b f(x)\, dx$. Step 1 · Express as a formal limit: $\int_1^{\infty} \frac{1}{x^2}\, dx = \lim \int_1^b x^{-2}\, dx$. Step 2 · Apply the Fundamental Theorem of Calculus: $\int_1^b x^{-2}\, dx = [-x^{-1}]_1^b = [-\frac{1}{x}]_1^b = \left(-\frac{1}{b}\right) - \left(-\frac{1}{1}\right) = 1 - \frac{1}{b}$. Step 3 · Evaluate the limit as $b \to \infty$: $\lim_{b\to\infty}\left(1 - \frac{1}{b}\right) = 1 - 0 = \mathbf{1}$. Conclusion: The improper integral converges to $\mathbf{1}$ (consistent with the p-integral test with $p = 2 > 1$).

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Euler's Method Iterations

Given $\frac{dy}{dx} = x + y$ and initial condition $y(0) = 1$, use Euler's method with step size $\Delta x = 0.5$ to approximate $y(1)$.

a. 2.25 ✓

b. 2.00

c. 2.50

d. 2.75

Step 1: at $(0, 1)$, slope $= 0 + 1 = 1$. $\Delta y = 1(0.5) = 0.5 \Rightarrow y(0.5) \approx 1.5$. Step 2: at $(0.5, 1.5)$, slope $= 0.5 + 1.5 = 2$. $\Delta y = 2(0.5) = 1.0 \Rightarrow y(1) \approx 1.5 + 1.0 = 2.25$.

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22 [Foundational] • Target: Euler's Method Iterations

If $y'' > 0$ (concave up) on the interval, does Euler's method yield an overestimate or underestimate of the true solution?

a. An underestimate because tangent segments lie below the curve ✓

b. An overestimate because tangent segments lie above the curve

c. Exact value

d. Depends on the step size

Euler's method uses tangent line segments. When the solution curve is concave up ($y'' > 0$), the curve bends above the tangent segments, making Euler's method an underestimate.

23 [Foundational] • Target: Euler's Method Iterations

Given $\frac{dy}{dx} = 2x$ with $y(1) = 3$, what is the error in approximating $y(2)$ using Euler's method with $\Delta x = 0.5$?

a. 0.5 ✓

b. 1.0

c. 0

d. 0.25

Euler: $y(1.5) = 3 + 2(1)(0.5) = 4$. $y(2) = 4 + 2(1.5)(0.5) = 4 + 1.5 = 5.5$. Exact: $y = x^2 + 2 \Rightarrow y(2) = 6$. Error $= 6 - 5.5 = 0.5$.

24 [Foundational] • Target: The Logistic Differential Equation

A population grows according to $\frac{dP}{dt} = 0.04P(1 - \frac{P}{500})$. What is the carrying capacity of this population?

a. 500 ✓

b. 0.04

c. 250

d. $12,500$

In the standard logistic equation $\frac{dP}{dt} = kP(1 - \frac{P}{M})$, the parameter M is the carrying capacity, which is 500.

25 [Foundational] • Target: The Logistic Differential Equation

For the logistic model $\frac{dP}{dt} = 0.04P(1 - \frac{P}{500})$, at what population size is the growth rate $\frac{dP}{dt}$ maximized?

a. 250 ✓

b. 500

c. 125

d. 375

The rate of growth is maximized at the point of inflection of the logistic curve, which occurs at exactly half the carrying capacity: $\frac{M}{2} = \frac{500}{2} = 250$.

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26 [Foundational] • Target: The Logistic Differential Equation

What is $\lim_{t\to\infty} P(t)$ for a logistic population with carrying capacity 800 if $P(0) = 50$?

a. 800 ✓

b. ∞

c. 400

d. 50

For any positive initial population $P_0 > 0$, the logistic curve asymptotically approaches the carrying capacity $M = 800$ as $t \to \infty$.

27 [Foundational] • Target: Analytical Solutions to Logistic Models

What is the general form of the solution to the logistic differential equation $\frac{dP}{dt} = kP(1 - \frac{P}{M})$?

a. $P(t) = \frac{M}{1 + Ae^{-kt}}$ ✓

b. $P(t) = Me^{kt}$

c. $P(t) = M - Ae^{-kt}$

d. $P(t) = \frac{M}{1 - Ae^{kt}}$

By separation of variables and partial fractions, the logistic solution is $P(t) = \frac{M}{1 + Ae^{-kt}}$.

28 [Foundational] • Target: Analytical Solutions to Logistic Models

If $P(t) = \frac{1000}{1 + 9e^{-0.2t}}$, what is the initial population $P(0)$?

a. 100 ✓

b. 1000

c. 90

d. 10

$P(0) = \frac{1000}{1 + 9(1)} = \frac{1000}{10} = 100$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Analytical Solutions to Logistic Models

What is the concavity of the logistic curve when $P(t) > \frac{M}{2}$?

a. **Concave down ($P''(t) < 0$)** ✓

b. Concave up ($P''(t) > 0$)

c. Zero concavity

d. Undefined

For $P < M/2$, the curve is concave up (accelerating growth). For $P > M/2$, the curve is concave down (decelerating growth).

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30 [Intermediate] • Target: Advanced Slope Fields with Autonomous Equations

In the autonomous differential equation $\frac{dy}{dx} = y(y-3)(y+2)$, what are all equilibrium solutions?

a. $y = 0, y = 3, y = -2$ ✓

b. $y = 3, y = -2$ only

c. $y = 0$ only

d. $x = 0, x = 3, x = -2$

Equilibrium solutions occur where $\frac{dy}{dx} = 0$, giving horizontal lines $y = 0, y = 3,$ and $y = -2$.

31 [Intermediate] • Target: Advanced Slope Fields with Autonomous Equations

For $\frac{dy}{dx} = 2y(1 - \frac{y}{4})$, which equilibrium solution is stable (an attractor)?

a. $y = 4$ ✓

b. $y = 0$

c. Both are stable

d. Neither is stable

Solutions starting near $y = 4$ approach 4 as $x \to \infty$ (since $y' > 0$ for $0 < y < 4$ and $y' < 0$ for $y > 4$), making $y = 4$ a stable attractor.

32 [Intermediate] • Target: Advanced Slope Fields with Autonomous Equations

If $y(0) = 6$ in $\frac{dy}{dx} = 2y(1 - \frac{y}{4})$, what is $\lim_{x \to \infty} y(x)$?

a. 4 ✓

b. 6

c. ∞

d. 0

Since $y(0) = 6 > 4$, $y' < 0$, so $y(x)$ decreases asymptotically toward the stable carrying capacity 4.

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33 [Intermediate] • Target: Euler's Method Iterations

A population of bacteria is modeled by the logistic differential equation $\frac{dP}{dt} = kP(1 - \frac{P}{L})$, where $P(t)$ is the population at time t in hours, k is the growth rate constant, and L is the carrying capacity. It is observed that $\frac{dP}{dt} = 200$ when $P = 100$, and the maximum growth rate occurs when $P = 500$. Find the carrying capacity L of this population.

a. 1000 ✓

b. 750

c. 1200

d. 500

The logistic differential equation is given by $\frac{dP}{dt} = kP(1 - \frac{P}{L})$. The growth rate $\frac{dP}{dt}$ is maximized when the population P is half of the carrying capacity L. We are given that the maximum growth rate occurs when $P = 500$. Therefore, $L/2 = 500$, which implies $L = 1000$. We can also use the other piece of information to verify this. When $P = 100$, $\frac{dP}{dt} = 200$. Substituting $P = 100$ and $L = 1000$ into the differential equation: $200 = k(100)(1 - \frac{100}{1000})$ $200 = 100k(1 - 0.1)$ $200 = 100k(0.9)$ $200 = 90k$ $k = \frac{200}{90} = \frac{20}{9}$. Now we check the maximum growth rate at $P = L/2 = 1000/2 = 500$ with $k = 20/9$ and $L = 1000$: $\frac{dP}{dt} = \frac{20}{9}(500)(1 - \frac{500}{1000})$ $\frac{dP}{dt} = \frac{10000}{9}(1 - 0.5)$ $\frac{dP}{dt} = \frac{10000}{9}(0.5)$ $\frac{dP}{dt} = \frac{5000}{9}$. This calculation confirms that our value of L = 1000 is correct, as the problem states that maximum growth rate occurs at P=500, and the differential of the logistic function is indeed maximized at P=L/2.

34 [Intermediate] • Target: The Logistic Differential Equation

Consider the logistic differential equation $\frac{dP}{dt} = kP(M - P)$, where $P(t)$ is the population at time t, k is the growth rate constant, and M is the carrying capacity. If $\frac{dP}{dt} = 0.05P(1000 - P)$, what is the carrying capacity of the population?

a. 1000 ✓

b. 0.05

c. 50

d. 0.00005

The logistic differential equation is given by $\frac{dP}{dt} = kP(M - P)$. In the given equation, $\frac{dP}{dt} = 0.05P(1000 - P)$. By comparing the two forms, we can identify the carrying capacity M. We see that $k = 0.05$ and $M = 1000$. Therefore, the carrying capacity of the population is 1000.

35 [Intermediate] • Target: Analytical Solutions to Logistic Models

Logistic growth model is $\frac{dP}{dt} =$

a. kP

b. $kP(1 - P/M)$ ✓

c. $k(M - P)$

d. P^2

Logistic: $kP(1 - P/M)$ where M is carrying capacity.

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36 [Intermediate] • Target: Advanced Slope Fields with Autonomous Equations

A colony grows according to $\frac{dP}{dt} = 0.11P\left(1 - \frac{P}{1100}\right)$. Find the carrying capacity and the population at which growth is fastest.

a. **M=1100,\ P=550 ✓**

b. M=1100,\ P=1100

c. M=550,\ P=1100

d. M=2200,\ P=550

Carrying capacity $M = 1100$; growth rate is maximal at the inflection point $P = M/2 = 550$.

37 [Intermediate] • Target: Euler's Method Iterations

A colony grows according to $\frac{dP}{dt} = 0.44P\left(1 - \frac{P}{440}\right)$. Find the carrying capacity and the population at which growth is fastest.

a. M=880,\ P=220

b. **M=440,\ P=220 ✓**

c. M=440,\ P=440

d. M=220,\ P=440

Carrying capacity $M = 440$; growth rate is maximal at the inflection point $P = M/2 = 220$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: The Logistic Differential Equation • Cross-Lesson: 6.02 Taylor and Maclaurin Series

A colony grows according to $\frac{dP}{dt} = 0.66P\left(1 - \frac{P}{660}\right)$. Find the carrying capacity and the population at which growth is fastest.

a. M=660,\ P=660

b. **M=660,\ P=330 ✓**

c. M=1320,\ P=330

d. M=330,\ P=660

Carrying capacity $M = 660$; growth rate is maximal at the inflection point $P = M/2 = 330$.

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39 [Challenging] • Target: Analytical Solutions to Logistic Models • Cross-Lesson: 1.01 Integration by Parts

A colony grows according to $\frac{dP}{dt} = 0.88P\left(1 - \frac{P}{880}\right)$. Find the carrying capacity and the population at which growth is fastest.

a. **M=880,\ P=440 ✓**

b. M=440,\ P=880

c. M=1760,\ P=440

d. M=880,\ P=880

Carrying capacity $M = 880$; growth rate is maximal at the inflection point $P = M/2 = 440$.

40 [Challenging] • Target: Advanced Slope Fields with Autonomous Equations • Cross-Lesson: 1.02 Partial Fractions and Improper Integrals

A colony grows according to $\frac{dP}{dt} = 0.33P\left(1 - \frac{P}{330}\right)$. Find the carrying capacity and the population at which growth is fastest.

a. M=660,\ P=165

b. M=330,\ P=330

c. M=165,\ P=330

d. **M=330,\ P=165 ✓**

Carrying capacity $M = 330$; growth rate is maximal at the inflection point $P = M/2 = 165$.

41 [Challenging] • Target: Euler's Method Iterations • Cross-Lesson: 3.01 Parametric Equations

A colony grows according to $\frac{dP}{dt} = 0.22P\left(1 - \frac{P}{220}\right)$. Find the carrying capacity and the population at which growth is fastest.

a. M=220,\ P=220

b. **M=220,\ P=110 ✓**

c. M=440,\ P=110

d. M=110,\ P=220

Carrying capacity $M = 220$; growth rate is maximal at the inflection point $P = M/2 = 110$.

42 [Challenging] • Target: The Logistic Differential Equation • Cross-Lesson: 3.02 Vector-Valued Functions

For $\frac{dP}{dt} = 0.05P\left(1 - \frac{P}{2000}\right)$, find the instantaneous growth rate when $P = 1000$.

a. -25

b. 25.5

c. **25 ✓**

d. 50

Substitute $P = 1000$: $0.05 \times 1000 \times \left(1 - \frac{1000}{2000}\right) = 25$.

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43 [Challenging] • Target: Analytical Solutions to Logistic Models • Cross-Lesson: 4.01 Polar Curves and Derivatives

For $\frac{dP}{dt} = 0.25P\left(1 - \frac{P}{800}\right)$, find the instantaneous growth rate when $P = 300$.

a. 93.75

b. -46.875

c. **46.875** ✓

d. 75

Substitute $P = 300$: $0.25 \times 300 \times \left(1 - \frac{300}{800}\right) = 46.875$.

44 [Challenging] • Target: Advanced Slope Fields with Autonomous Equations • Cross-Lesson: 4.02 Area in Polar Coordinates

For $\frac{dP}{dt} = 0.55P\left(1 - \frac{P}{350}\right)$, find the instantaneous growth rate when $P = 140$.

a. **46.2** ✓

b. -46.2

c. 92.4

d. 77

Substitute $P = 140$: $0.55 \times 140 \times \left(1 - \frac{140}{350}\right) = 46.2$.

45 [Challenging] • Target: Euler's Method Iterations • Cross-Lesson: 5.01 Sequences and Basic Series Convergence Tests

For $\frac{dP}{dt} = 0.85P\left(1 - \frac{P}{60}\right)$, find the instantaneous growth rate when $P = 20$.

a. **11.3333** ✓

b. 17

c. 22.6667

d. -11.3333

Substitute $P = 20$: $0.85 \times 20 \times \left(1 - \frac{20}{60}\right) = 11.3333$.

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